Research Papers: Urban Flood Nowcasting System

(Drainage and Rainfall Coupling) | SIH 2026, SIH26085 | Ministry of Earth Sciences

25 papers grouped by system component. Links are clickable. † = some bibliographic details (full author list, venue or year) could not be fully confirmed; verify on Google Scholar or via the DOI before citing.

A. Rainfall Nowcasting (Radar and Deep Learning)

1. Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., Woo, W.-C. (2015). *Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting.* NeurIPS 28.

   Relevance: Introduces ConvLSTM, framing nowcasting as spatiotemporal sequence forecasting.

   https://proceedings.neurips.cc/paper/2015/hash/07563a3fe3bbe7e3ba84431ad9d055af-Abstract.html
   https://arxiv.org/pdf/1506.04214

2. Ravuri, S., et al. (2021). *Skilful precipitation nowcasting using deep generative models of radar.* Nature. DOI: 10.1038/s41586-021-03854-z.

   Relevance: Probabilistic generative nowcasting model ranked first by expert meteorologists in 89% of cases.

   https://doi.org/10.1038/s41586-021-03854-z
   https://pmc.ncbi.nlm.nih.gov/articles/PMC8481123

3. Zhang, Y., Long, M., Chen, K., Xing, L., Jin, R., Jordan, M.I., Wang, J. (2023). *Skilful nowcasting of extreme precipitation with NowcastNet.* Nature 619, 526-532.

   Relevance: Unifies physical-evolution schemes with conditional learning for extreme precipitation, up to 3 h lead time.

   https://www.nature.com/articles/s41586-023-06184-4

4. Gao, Z., Shi, X., Wang, H., Zhu, Y., Wang, Y., Li, M., Yeung, D.-Y. (2022). *Earthformer: Exploring Space-Time Transformers for Earth System Forecasting.* NeurIPS 35.

   Relevance: Space-time Transformer (Cuboid Attention) with state-of-the-art precipitation nowcasting results.

   https://proceedings.neurips.cc/paper_files/paper/2022/hash/a2affd71d15e8fedffe18d0219f4837a-Abstract.html
   https://arxiv.org/pdf/2207.05833

5. Pulkkinen, S., Nerini, D., Perez Hortal, A.A., Velasco-Forero, C., Seed, A., Germann, U., Foresti, L. (2019). *Pysteps: an open-source Python library for probabilistic precipitation nowcasting (v1.0).* Geosci. Model Dev. 12, 4185-4219.

   Relevance: Open-source 0-6 h probabilistic nowcasting library; a practical baseline for the prototype.

   https://gmd.copernicus.org/articles/12/4185/2019/

B. Nowcasting Coupled with Urban Drainage and Flood Models

6. Thorndahl, S., Nielsen, J.E., Jensen, D.G. (2016). *Urban pluvial flood prediction: a case study evaluating radar rainfall nowcasts and numerical weather prediction models as model inputs.* Water Sci. Technol. 74(11), 2599-2610.

   Relevance: Feeds radar, 0-2 h nowcasts and NWP forecasts into an integrated flood and drainage model.

   https://pubmed.ncbi.nlm.nih.gov/27973364/
   https://iwaponline.com/wst/article/74/11/2599/18948/Urban-pluvial-flood-prediction-a-case-study

7. Schellart, A., Ochoa, S., et al. (2011). *Urban pluvial flood modelling with real time rainfall information - UK case studies.* 12th Int. Conf. on Urban Drainage, Porto Alegre.

   Relevance: Rainfall forecasting plus drainage modelling for pluvial flood warning, incl. ANN-based RAPIDS.

   https://core.ac.uk/download/76991284.pdf

8. Codo, M., Rico-Ramirez, M.A. (2018). *Ensemble Radar-Based Rainfall Forecasts for Urban Hydrological Applications.* Geosciences 8(8), 297. DOI: 10.3390/geosciences8080297.
Relevance: Radar rainfall forecast ensembles used to predict sewer flows (InfoWorks CS).
https://core.ac.uk/download/160810581.pdf

9. Liguori, S., Rico-Ramirez, M.A., Schellart, A.N.A., Saul, A. (c. 2012). *Using probabilistic radar rainfall nowcasts and NWP forecasts for flow prediction in urban catchments.* (†)
Relevance: Probabilistic nowcasts and NWP forecasts for urban flow prediction (University of Bristol).
https://research-information.bris.ac.uk/en/publications/using-probabilistic-radar-rainfall-nowcasts-and-nwp-forecasts-for/

10. Wang, L., et al. (2011). *Radar-based pluvial flood forecasting over urban areas: Redbridge case study.* IAHS Publ., Weather Radar and Hydrology. (†)
Relevance: Couples the UK Met Office STEPS nowcasting model with an urban drainage model.
https://core.ac.uk/download/76991275.pdf

11. Ochoa-Rodriguez, S., Rico-Ramirez, M.A., Jewell, S.A., Schellart, A.N.A., et al. *Improving rainfall nowcasting and urban runoff forecasting through dynamic radar-raingauge rainfall adjustment.* (†)
Relevance: Gauge-adjusted radar nowcasts fed into a London storm-water drainage model.
https://core.ac.uk/download/76991268.pdf

C. ML Surrogates and Fast Flood Prediction

12. McSpadden, D., Goldenberg, S., Roy, B., Schram, M., Goodall, J.L., Richter, H. (2024). *A comparison of machine learning surrogate models of street-scale flooding in Norfolk, Virginia.* Machine Learning with Applications 15, 100518. DOI: 10.1016/j.mlwa.2023.100518.
Relevance: RNN, LSTM and GRU surrogates for real-time street-scale flood prediction.
https://arxiv.org/abs/2307.14185v1
https://doi.org/10.1016/j.mlwa.2023.100518

13. Zahura, F.T., Goodall, J.L., et al. (2020). *Training machine learning surrogate models from a high-fidelity physics-based model: application for real-time street-scale flood prediction in an urban coastal community.* Water Resources Research. (†)
Relevance: Random forest surrogate cut physics-model run time by a factor of 3,000. Link is an author listing page; locate the paper title there.
https://par.nsf.gov/search/author:%22Morsy,%20Mohamed_M%22

14. Asif, M., Albano, R. (2026). *Convolutional neural network model for rapid prediction of urban flash flood water depth and velocity maps.* (†)
Relevance: CNN surrogate predicting depth and velocity from a rainfall hyetograph and hydro-morphological descriptors.
https://iris.unibas.it/handle/11563/214897

15. Tyagi, P., Dubey, V. (2025). *Urban Flood-Depth Forecasting with Autoregressive Surrogates: A Case Study of Indian Cities.* BMVC 2025 workshop.
Relevance: ML flood-depth forecasts validated on the 2005 Mumbai and 2015 Chennai floods.
https://bmva-archive.org.uk/bmvc/2025/assets/workshops/MVCC/Paper_3/paper.pdf

16. Lou, W., Gao, X., Lee, J.H.W., Liu, J., Dong, L., Gao, K. (2026). *A highly generalizable data-driven model for spatiotemporal urban flood dynamics real-time forecasting based on coupled CNN and ConvLSTM.* Hydrol. Earth Syst. Sci. 30, 1625-1646. DOI: 10.5194/hess-30-1625-2026.
Relevance: Generalizable CNN + ConvLSTM real-time urban flood forecasting model.
https://hess.copernicus.org/articles/30/1625/2026/

D. Physics-Based Surface and Drainage Coupling

17. (2026). *A computationally efficient surface-drainage network coupled hydrodynamic model for urban flood simulation.* Water Practice & Technology 21(3), 1166-1187. DOI: 10.2166/wpt.2026.237. (†)
    Relevance: Bidirectionally couples a GPU-accelerated surface runoff model with a drainage network model; local time stepping gives up to 9.35x speedup.
    https://iwaponline.com/wpt/article-abstract/21/3/1166/111149/A-computationally-efficient-surface-drainage?redirectedFrom=fulltext

18. (2024). *Urban flood modeling with a novel coupling method of surface and sewer hydrodynamic processes.* Water Sci. Technol. 89(11), 3021. (†)
    Relevance: Combines 2D-1D and 1D-2D unidirectional processes to overcome limits of conventional one-way coupling.
    https://iwaponline.com/wst/article/89/11/3021/102436/Urban-flood-modeling-with-a-novel-coupling-method

19. (2023). *Analysis of regional flooding in the urbanization expansion process based on the SWMM model.* Natural Hazards 117(2), 1349-1363. DOI: 10.1007/s11069-023-05906-1. (†)
    Relevance: SWMM storm flood models for different recurrence periods in Xinxiang.
    https://ideas.repec.org/a/spr/nathaz/v117y2023i2d10.1007_s11069-023-05906-1.html

20. *Rainfall-runoff modeling of urban floods using GIS and HEC-HMS.* (PMC12226067). (†)
    Relevance: Uses a 7 m DEM with SCS-CN, SCS-UH and Muskingum routing, with calibration.
    https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12226067/

E. Indian Urban Flood Forecasting Case Studies

21. Ghosh, S., et al. (2019). *Development of India's first integrated expert urban flood forecasting system for Chennai.* Current Science 117(5), 741. (†)
    Relevance: Couples weather forecast, tide, tidal flood, overland flow and storm-water drainage models; led by IIT Bombay with MoES partners.
    https://services.incois.gov.in/documents/ResearchPapers/RP118.pdf

22. Ghosh, M., Paul, S., Karmakar, S., Ghosh, S. (2021). *Near-real-time flood forecasting for an urban coastal catchment: combining numerical weather and 3-way coupled hydrodynamic flood modelling.* EGU General Assembly, EGU21-12834.
    Relevance: WRF rainfall forecasts feeding MIKE 11 / MIKE 21 flood models for Mumbai.
    https://meetingorganizer.copernicus.org/EGU21/EGU21-12834.html

23. Mujumdar, P.P., Mohan Kumar, M.S., Reddy, G.S.S., Avinash, S., Chawla, I., Kaushika, G.S., Parashuram, K., Kumar, A. (2021). *Development of an urban flood model for Bengaluru city, Karnataka, India.* Current Science 120(9), 1441-1448.
    Relevance: Modular system: real-time rainfall monitoring, rainfall forecast, and a SWMM-based flood model.
    https://eprints.iisc.ac.in/69520/

24. *Identification of Critical Hotspots in Urban Drainage Networks using MIKE URBAN.* CHI Journal (Rohtak, Haryana case study). (†)
    Relevance: MIKE+ 1D hydrodynamic model used to identify hotspot nodes in the 2022 flood event.
    https://www.chijournal.org/H539

25. Kelkar, V., Solanki, V., Krebs, P. *Blue-green infrastructure and urban flood reduction, F-North and G-North wards of Greater Mumbai.* Water 18, 1542. (†)
    Relevance: QGIS + EPA-SWMM modelling under a 25-year return period rainfall scenario.
    https://mdpi-res.com/d_attachment/water/water-18-01542/article_deploy/water-18-01542.pdf